

Hayden Rivas

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About Me

College student enthusiastic about connecting with others and thriving in a collaborative environment. Currently, pivoting my deep experience in graphics programming and low-level optimizations into a broader software development role.

Experience

QuackHacks UO (Jan 2026 – Present)

Instagram [🔗](#)

- Led UI/UX design and frontend development for a hackathon platform that onboarded **500+ applicants**, architecting the landing page, admin panel, and user dashboard using **Next.js**, **React**, and **TailwindCSS**, delivering the full build over a **10-week** term.
- Engineered an immersive 3D experience with **Three.js** and enforced end-to-end type safety via **TypeScript**, producing a polished, production-quality interface for a **24-hour** live hackathon event.
- Architected real-time backend pipelines through **Supabase** to handle **250+ concurrent users**, ensuring a seamless participant onboarding flow with zero downtime during the live event.

Projects

Pimp My Canvas

Chrome Web Store Page [🔗](#)

- Engineered an official Chrome Extension, featured on the front page, with **680k+ impressions** across Chrome and **15k+ active users**.
- Leveraged **HTML/CSS** alongside **Bootstrap** to create a custom styling engine, allowing users to personalize their web experience with custom themes across the Canvas LMS.

Mythril - Vulkan Framework

externalhost0/Mythril [🔗](#)

- Developed a high-performance, general purpose graphics library using **C++ 20** to provide a clean abstraction layer over the **Vulkan** API.
- Integrated the **Slang** shading language for programmable pipeline support and utilized **CMake** for a modular, cross-platform build system.

ts-play-core

externalhost0/ts-play-core [🔗](#)

- Authored and published an official **NPM** package in **TypeScript** that enables developers to write shader-like logic for rendering complex ASCII art and animations on the web, accumulating **750+ downloads** within its first release cycle.
- Designed a typesafe API surface that abstracts real-time procedural text generation, reducing integration complexity for consuming developers and earning organic traction with **zero marketing spend**.

spotty.nvim

externalhost0/spotty.nvim [🔗](#)

- To eliminate context-switching during development sessions, engineered a lightweight **Lua** Neovim plugin integrating the **Spotify Web API** to deliver real-time playback metadata directly in the editor without disrupting the editing experience.
- Solved the challenge of non-blocking I/O in a single-threaded editor environment by implementing asynchronous **curl** requests and custom UI components via the Neovim API, achieving zero impact on editor performance.

Slated_Vulkan

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- Designed and built a cross-platform ECS game engine from scratch with a custom **Vulkan** renderer, a declarative renderpass system, and an extensible resource pipeline, demonstrating low-level graphics and engine architecture at a production-adjacent scale.
- Architected a cross-language communication layer using sockets and IPC to synchronize state and stream data between a **C++17** engine core and a **C#** editor frontend, enabling a seamless hybrid editor workflow across two runtimes.

Education

University of Oregon, Eugene, OR

September 2024 - Present

- Computer Science Major, CS GPA: 3.7
- Coursework: Intermediate Data Structures, Operating Systems, Discrete Mathematics, Linear Algebra

Technologies

Languages: C++, C, Python, Lua, JavaScript, TypeScript, GLSL/HLSL/Slang, HTML/CSS, SQL

Frameworks & Layers: Vulkan, OpenGL, React, Next.js, Svelte, Astro, Deno, Supabase, Node, jQuery, Tailwind CSS,

Developer Tools: Git, CMake, Ninja, RenderDoc, GitHub Actions, Docker, GDB, LaTeX, Bootstrap